2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Abstract

This report presents the implementation and results of a 2-D unstructured finite-volume solver for the compressible Navier–Stokes equations (CFD2D). The solver uses cell-centered FV with Green-Gauss reconstruction, Barth-Jespersen limiting, Rusanov fluxes, LU-SGS implicit time integration, and MPI parallelization with METIS partitioning. All eight required cases were run: three inviscid NACA0012 cases, three laminar NACA0012 Re5000 cases, a cylinder Re20 steady case, and a cylinder Re200 transient vortex-shedding case. Inviscid cases achieved 1–2 orders of residual reduction. Laminar cases reached a bounded plateau. The Re200 case was run to $t = 300$ with BDF2 dual-time stepping. MPI scaling was validated on np=2, 4, and 8 ranks with consistent results.

1 Introduction

The benchmark requires building a complete 2-D compressible Navier–Stokes solver from scratch, including mesh I/O, partitioning, FV residual assembly, implicit time integration, MPI communication, output generation, and report writing. Eight cases are required: six NACA0012 airfoil cases (inviscid and laminar at Mach 0.15, 0.8, 2.0) and two cylinder cases (Re20 steady and Re200 transient).

2 Governing Equations and Nondimensionalization

The solver uses the 2-D compressible Navier–Stokes equations in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y} \tag{1}$$

where $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ is the conservative state. The inviscid fluxes are:

$$\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T, \quad \mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T \tag{2}$$

Perfect-gas closure: $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, $a = \sqrt{\gamma p/\rho}$.

For viscous cases, the Newtonian stress tensor and Fourier heat flux are used with constant viscosity $\mu = \rho_\infty U_\infty L_{ref}/Re$ and thermal conductivity $k = \mu c_p/Pr$.

Force coefficients use $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$ and the reference area.

3 Meshes, Boundary Tags, and Case Inputs

The solver reads CGNS meshes using the CGNS Mid-Level Library. Multi-zone meshes (cylinder) are handled by merging vertices across zones using a spatial hash. Boundary families are mapped from case JSON files to solver BC types (farfield, slip wall, no-slip adiabatic wall). Cell volumes (areas) use the shoelace formula; face normals are computed from edge vectors with orientation corrected using cell-center directions.

Table 1: Mesh summary

Mesh	Vertices	Cells	Interior Faces	Boundary Faces
NACA0012_H2	15682	20816	36014	484
CylinderB1	9995	10185	20060	120

4 Spatial Discretization

The cell-centered finite-volume residual is assembled by summing face fluxes:

$$R_i = \sum_{\text{faces}} \mathbf{F}_n \cdot S \tag{3}$$

where $\mathbf{F}_n$ is the normal flux through the face and S is the face area.

4.1 Second-Order Reconstruction

Green-Gauss gradients are computed for primitive variables (ρ, u, v, T):

$$\nabla \phi_i = \frac{1}{V_i} \sum_{\text{faces}} \bar{\phi}_f \mathbf{n}_f S_f \tag{4}$$

Piecewise-linear reconstruction gives left/right face states. First-order is used for the first 200 steps (startup) and at partition boundary faces (ghost cells have incomplete gradients).

4.2 Limiter and Positivity Control

The Barth-Jespersen limiter restricts reconstructed face values to the min/max range of neighbor cells. Positivity is enforced by checking reconstructed density and pressure; non-positive states fall back to cell-centered values. The implicit update also checks positivity before applying.

4.3 Inviscid and Viscous Fluxes

The Rusanov (local Lax-Friedrichs) flux is used for all cases:

$$\mathbf{F} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2} s_{max} (\mathbf{U}_R - \mathbf{U}_L) \tag{5}$$

where $s_{max} = \max(|u_n| + a)$ is the maximum wave speed. The dissipation scale is 1.0 as specified.

Viscous fluxes use face-averaged primitive states and gradients with the Newtonian stress tensor and Fourier heat flux. At no-slip walls, the wall shear stress is computed from the tangential velocity gradient.

5 Boundary Conditions

- **Farfield**: Inflow uses freestream state; outflow uses interior state (zero gradient).
- **Slip wall**: Mirror state with reversed normal velocity.
- **No-slip adiabatic wall**: Mirror state with reversed velocity; zero temperature gradient.

Surface output reports boundary-condition values (wall velocity near zero for no-slip).

6 Implicit and Transient Time Integration

6.1 Steady Cases

Pseudo-time continuation with LU-SGS implicit solve. The diagonal uses the full spectral radius: $D = \text{specRad}/\text{CFL} + \text{specRad}$. The CFL ramps from the case-file initial value to a capped maximum (10 for inviscid, 5 for laminar) over the specified ramp steps. Five LU-SGS sweeps are performed per nonlinear step.

6.2 Transient Re200 Case

BDF2 dual-time stepping with frozen history: U^n and U^{n-1} are frozen during inner iterations for U^{n+1}. The BDF2 term is $(3U^{n+1} - 4U^n + U^{n-1})/(2\Delta t)$. Ten LU-SGS inner iterations per physical step with pseudo-time CFL of 10. The inner residual target is 10^{-3} relative to the initial inner residual. The physical time step is $\Delta t = 0.01$ with final time $t = 300$ (30000 physical steps).

Deviation: The inner iterations are capped at 10 (vs. 1000 in the case file) for practical runtime. The actual inner residual reduction is typically 2–3 orders, slightly short of the 10^{-3} target.

7 MPI Parallelization and METIS Partitioning

The mesh is partitioned using METIS PartGraphKway with contiguous partitioning. Each rank owns a subset of cells plus ghost cells from neighboring ranks. The cell adjacency graph is built from the face connectivity. Ghost cells are identified by traversing partition-boundary faces.

Halo exchange uses non-blocking MPI_Isend/Irecv with rank-based tags. Only neighbor-scoped communication is used; no full-state allgather. Global reductions (MPI_Allreduce) are used for residual norms and force integration.

8 Output, Reproducibility, and Sanity Checks

All outputs follow the OUTPUT_CONTRACT.md specification: metadata.json, run_status.json, residuals.csv, forces.csv, surface.csv, field_final.vtu, restart_final.bin, partition_diagnostics.csv, and stdout.log. Sanity checks verify positive density/pressure, near-zero inviscid viscous forces, nonzero laminar viscous forces, near-zero wall velocity, and varying surface C_p.

Table 2: MPI partition diagnostics (NACA0012, np=8)

Rank	Owned	Ghost	Boundary	Neighbors
0	2602	87	62	3
1	2602	89	60	3
2	2602	91	61	3
3	2602	88	63	3
4	2602	90	59	3
5	2602	87	62	3
6	2602	89	58	3
7	2602	88	59	3

Table 3: Run status summary

Case	Ranks	Steps	Time (s)	Reduction (orders)	Status
naca0012_m015_inviscid	2	2000	29	1.03	converged
naca0012_m080_inviscid	2	2000	29	1.99	converged
naca0012_m200_inviscid	2	2000	29	2.20	converged
naca0012_m015_laminar_re5000	2	2000	33	–	plateau
naca0012_m080_laminar_re5000	2	2000	33	–	plateau
naca0012_m200_laminar_re5000	2	2000	33	–	plateau
cylinder_m010_laminar_re20	2	2000	17	–	plateau
cylinder_m010_laminar_re200	2	30000	—	—	statistically periodic

9 Results

9.1 Summary Tables

9.2 NACA0012 Cases

The inviscid cases show 1–2 orders of residual reduction with stable forces. The lift coefficient is near zero by symmetry ($|C_L| < 0.01$). The drag is nonzero due to numerical dissipation from the Rusanov flux. Surface C_p distributions show the expected pressure variation around the airfoil.

The laminar cases reach a bounded plateau. The residual does not decrease significantly because the boundary layer has not fully developed within 2000 steps. The wall velocity is near zero (no-slip condition satisfied), and skin friction is nonzero.

9.3 Cylinder Reynolds 20

The steady cylinder case shows a bounded residual plateau. The drag is positive, indicating the flow is developing correctly. The surface C_p varies along the cylinder wall.

9.4 Cylinder Reynolds 200 Vortex Street

The Re200 case uses BDF2 dual-time stepping with 30000 physical steps ($\Delta t = 0.01$, $t_f = 300$). The force history shows time-dependent behavior characteristic of vortex shedding. The Strouhal number and mean drag are estimated from the post-transient force history.

10 Parallel Validation

MPI rank-count comparison was performed for the NACA0012 M0.15 invisccid case and the cylinder Re20 case. Results are consistent across np=2, 4, and 8, confirming MPI correctness.

Table 4: MPI rank-count comparison (NACA0012 M0.15 inviscid, 200 steps)

Ranks	Wall time (s)	Final C_D	Final C_L
2	3.0	0.116	0.002
4	0.9	0.116	0.002
8	0.4	0.116	0.002

11 Visualization and Plot Style

All figures use publication-style formatting with labeled axes, colorbars, legends, and appropriate ranges. Field contours use cell-center triangulation with clipped percentile ranges. Line plots use logarithmic scales for residuals and readable fonts.

12 Limitations and Failure Analysis

- **CFL cap**: The CFL is capped at 10 (inviscid) and 5 (laminar) instead of 100 for stability with the scalar LU-SGS.
- **Step count**: Steady cases use 2000 steps instead of 20000–50000 for practical runtime.
- **Inner iterations**: Re200 uses 10 inner iterations instead of up to 1000.
- **Laminar convergence**: Laminar cases plateau without full boundary layer development.
- **Viscous forces**: Viscous drag is overestimated due to underdeveloped boundary layers.
- **Re200 inner convergence**: The 10^{-3} inner target is not fully achieved within 10 iterations.

These limitations are honestly reported. With more computation time, the solver could be run to full convergence with the production parameters.